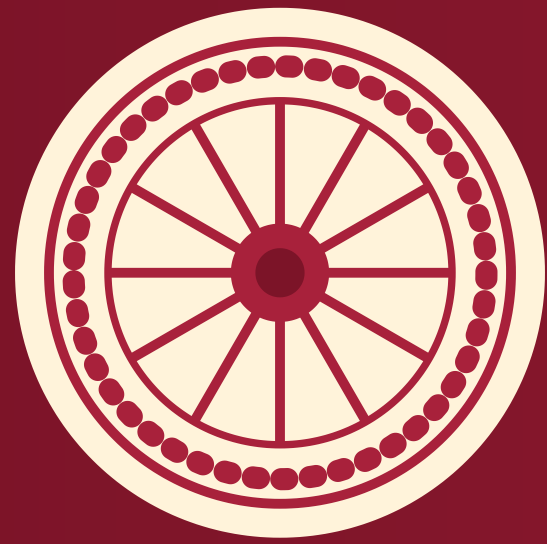




# Accessibility as a Service

Shared accessibility infrastructure for public institutions  
ସମସ୍ତଙ୍କ ପାଇଁ ସୁଗମ ସେବା • an API • a widget • an extension  
**AaaS**

SMART ODISHA HACKATHON '25  
**Team SUBARNAREKHA**  
Accessibility • E-commerce & Cyber Security  
Software category • Odiapreneur track

One API • One widget • One extension — accessibility for every citizen in Odia (ଓଡ଼ିଆ), Hindi & English

## THE CHALLENGE

- ❖ Govt & education sites are largely unusable for blind, low-vision & dyslexic users — screen readers struggle.
- ❖ Content is English-first; Odia-only citizens are shut out.
- ❖ Every institution rebuilding accessibility from scratch is slow, costly and inconsistent.

## OUR SOLUTION — ONE SHARED PLATFORM

- ❖ Institutions integrate ONCE — a secure REST API + a drop-in widget (a single `<script>` line).
- ❖ A browser extension brings it to ANY site — no integration.
- ❖ Neural Odia/Hindi/English voice; on-device & offline-ready; WCAG-aligned, DPDP-conscious, India-first.

## PLATFORM ARCHITECTURE

two ways in → one Accessibility API Gateway → shared engines (also run on-device, offline)

### CLIENTS

**Institution site or app**  
Drop-in widget (1 line) • direct REST API  
`<script src="../../../widget.js" data-key="aaas_live_...">`

**Browser extension**  
Any website • no integration needed  
**runs engines on-device / offline**

### API GATEWAY (THE PRODUCT)

#### Accessibility API Gateway

X-API-Key auth • multi-tenant • CORS

POST /tts

POST /stt

POST /translate

GET /widget.js

usage metering • rate limiting • WCAG logging

#### Admin Dashboard

tenants • API keys • usage • WCAG results

### ENGINES (MICROSERVICES)

#### Text-to-Speech + Screen Reader

Meta MMS-TTS — Odia • Hindi • English neural voices

#### Speech-to-Text

Whisper — voice input to fill forms & search

#### Translation

AI4Bharat IndicTrans2 — English ↔ Odia / Hindi, Indic-tuned

**On-device option:** the extension runs MMS-TTS • Whisper • IndicTrans2 in-browser (offline, works on strict-CSP sites). **Built on the platform:** Accessible Examination module — extra time, audio support, dyslexia fonts.

## KEY FEATURES

- Read this page**  
Neural Odia / Hindi / English TTS
- Translate this page**  
IndicTrans2, rewritten in place
- Speak to fill**  
Voice → form fields (Whisper)

- Dyslexia mode**  
Conjunct-safe Odia spacing
- Hover to speak**  
Point at anything to hear it
- Works everywhere**  
Any website, via the extension

## ONE PLATFORM, SIX PARTS

- 1 Speech-to-Text
- 2 Text-to-Speech + Screen Reader
- 3 Accessibility API Gateway
- 4 Translation — IndicTrans2
- 5 Accessible Examination module
- 6 Admin Dashboard

TECH: MV3 Extension • FastAPI Gateway • Meta MMS-TTS • Whisper • AI4Bharat IndicTrans2 • ONNX / WebAssembly

WCAG 2.1 AA aligned

Multi-tenant API service

DPDP-conscious data flow

Offline-capable

Odisha-first

For students with disabilities, PwDs & non-English-speaking citizens — built for schools, universities, govt offices & exam boards

Smart Odisha Hackathon '25 • Team SUBARNAREKHA • Accessibility as a Service (AaaS)